

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1

Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I _S.Jotheeswaran(192424155)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.Jotheeswaran

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand
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Case Study 1: Smart Medical Diagnosis System

Theoretical Background

In Artificial Intelligence, a rule-based expert system uses a knowledge base (KB) consisting of a set of rules and facts to deduce new information. Propositional logic provides a foundational framework for representing these rules using propositions (statements that are either True or False) and logical connectives (AND, OR, NOT, IMPLIES). Inference mechanisms such as Forward Chaining and Backward Chaining are used to derive conclusions or verify goals against the KB.

(a) Representation in Propositional Logic

First, we define the propositional symbols used to represent the medical conditions and symptoms accurately:

- F: Patient has Fever
- C: Patient has Cough
- R: Patient has Rash
- B: Patient has Breathlessness
- Flu: Patient has Possible Flu
- Measles: Patient has Possible Measles
- Pneumonia: Patient has Possible Pneumonia

Using these symbols, the rules and facts from the Knowledge Base (KB) are formally translated as follows:

Rules:

Rule I: $F \wedge C \rightarrow \text{Flu}$

Rule II: $F \wedge R \rightarrow \text{Measles}$

Rule III: $C \wedge B \rightarrow \text{Pneumonia}$

Facts for Patient A:

F = True (Fever is present)

C = True (Cough is present)

B = True (Breathlessness is present)

(b) Applying Forward Chaining

Forward chaining is a data-driven inference method. We start with the known facts and iteratively apply rules whose premises (antecedents) are satisfied to deduce new facts, stopping when no more new facts can be derived.

Iteration 1:

- Current known facts: {F, C, B}

- Evaluating Rule I ($F \wedge C \rightarrow \text{Flu}$): Both F and C are in our set of known facts. The rule fires. New fact deduced: Flu.
- Evaluating Rule II ($F \wedge R \rightarrow \text{Measles}$): F is known, but R is not. The rule does not fire.
- Evaluating Rule III ($C \wedge B \rightarrow \text{Pneumonia}$): Both C and B are in our set of known facts. The rule fires. New fact deduced: Pneumonia.

Iteration 2:

- Current known facts: {F, C, B, Flu, Pneumonia}
- Evaluating remaining unfired Rule II: R is still not known. The rule does not fire.

Termination:

No additional rules can be fired. The forward chaining algorithm terminates. The derived diagnoses for Patient A are Possible Flu and Possible Pneumonia.

(c) Applying Backward Chaining (Goal Reduction)

Backward chaining is a goal-driven inference method. We start with a specific goal and work backward through the rules to see if the goal can be supported by the known facts.

Goal: Confirm 'Patient A has Pneumonia' (Symbol: Pneumonia)

Step-by-step execution:

1. Current Goal Stack: [Pneumonia]
2. Find a rule that concludes 'Pneumonia'. We find Rule III: $C \wedge B \rightarrow \text{Pneumonia}$.
3. To prove Pneumonia, we must now prove its prerequisites: C (Cough) and B (Breathlessness). New Goal Stack: [C, B]
4. Pop 'C' from the stack. Is 'C' a known fact? Yes, C = True is given in the KB. Goal 'C' is resolved.
5. Current Goal Stack: [B]
6. Pop 'B' from the stack. Is 'B' a known fact? Yes, B = True is given in the KB. Goal 'B' is resolved.
7. Since all sub-goals are satisfied, the original goal 'Pneumonia' is successfully confirmed.

Goal Reduction Tree:

[Pneumonia]

/ \

(Rule III)

/ \

[C] [B]

(Fact) (Fact)

True True

Case Study 2: Autonomous Traffic Management Agent

Theoretical Background

First-Order Logic (FOL) extends propositional logic by introducing predicates, variables, and quantifiers (Universal $\forall$, Existential $\exists$), allowing for more expressive knowledge representation. Unification is the process of finding a substitution (Most General Unifier, MGU) that makes two logical expressions identical. Resolution is a sound and complete inference rule used for proof by contradiction (refutation) in FOL, which requires converting sentences into Conjunctive Normal Form (CNF).

(a) Unification Process

Unification matches variables in a generalized rule to specific constants from our facts to extract actionable knowledge.

Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Given Facts:

F1: $\text{Vehicle}(\text{Ambulance})$

F2: $\text{EmergencyType}(\text{Ambulance})$

Unification Step:

To match the premise of Rule 1, we unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$.

This yields the substitution set θ (Most General Unifier, MGU): $\{x / \text{Ambulance}\}$.

Applying this MGU to the entire rule gives: $\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$. Since the premises are true facts, we derive $\text{ClearPath}(\text{Ambulance})$.

(b) Resolution Method to Prove Proceed(CarA)

Step 1: Convert all rules and facts to Conjunctive Normal Form (CNF).

Eliminate implications ($A \rightarrow B$ becomes $\neg A \vee B$) and distribute:

1. Rule 1: $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

2. Rule 2: $\neg \text{ClearPath}(y) \vee (\text{GreenSignal}(y) \wedge \neg \text{RedSignal}(y))$

- Split into two clauses:

2a. $\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$

2b. $\neg \text{ClearPath}(y) \vee \neg \text{RedSignal}(y)$

3. Rule 3: $\neg \text{Vehicle}(z) \vee \neg \text{Behind}(z, w) \vee \neg \text{GreenSignal}(w) \vee \text{Proceed}(z)$

4. Facts (already in CNF):

F1: Vehicle(Ambulance)

F2: EmergencyType(Ambulance)

F3: Vehicle(CarA)

F4: Behind(CarA, Ambulance)

Step 2: Negate the Goal

Goal: Proceed(CarA). Negated Goal: $\neg \text{Proceed}(\text{CarA})$. Add this to our knowledge base to derive a contradiction.

Step 3: Resolution Proof

Step R1: Resolve Negated Goal [$\neg \text{Proceed}(\text{CarA})$] with Rule 3 [$\neg \text{Vehicle}(z) \vee \neg \text{Behind}(z, w) \vee \neg \text{GreenSignal}(w) \vee \text{Proceed}(z)$].

Substitution: $\{z / \text{CarA}\}$

Resulting Resolvent: $\neg \text{Vehicle}(\text{CarA}) \vee \neg \text{Behind}(\text{CarA}, w) \vee \neg \text{GreenSignal}(w)$

Step R2: Resolve result with Fact [Vehicle(CarA)].

Resulting Resolvent: $\neg \text{Behind}(\text{CarA}, w) \vee \neg \text{GreenSignal}(w)$

Step R3: Resolve result with Fact [Behind(CarA, Ambulance)].

Substitution: $\{w / \text{Ambulance}\}$

Resulting Resolvent: $\neg \text{GreenSignal}(\text{Ambulance})$

Step R4: Resolve result with Rule 2a [$\neg \text{ClearPath}(y) \vee \text{GreenSignal}(y)$].

Substitution: $\{y / \text{Ambulance}\}$

Resulting Resolvent: $\neg \text{ClearPath}(\text{Ambulance})$

Step R5: Resolve result with Rule 1 [$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$].

Substitution: $\{x / \text{Ambulance}\}$

Resulting Resolvent: $\neg \text{Vehicle}(\text{Ambulance}) \vee \neg \text{EmergencyType}(\text{Ambulance})$

Step R6: Resolve result with Fact [EmergencyType(Ambulance)].

Resulting Resolvent: $\neg \text{Vehicle}(\text{Ambulance})$

Step R7: Resolve result with Fact [Vehicle(Ambulance)].

Resulting Resolvent: Empty Clause (Contradiction!)

Conclusion: Since deriving the empty clause means the negated goal contradicts the KB, the original goal `Proceed(CarA)` is unequivocally proven to be True.

(c) Analysis of Simultaneous Emergency Vehicles

Current KB limitation: The current First-Order Logic framework dictates that any vehicle designated as an `EmergencyType` receives a `GreenSignal`. If two emergency vehicles (e.g., `Ambulance1` and `Ambulance2`) approach the same intersection from perpendicular directions, applying the existing rules will independently grant a `GreenSignal` to both paths. In a physical traffic system, this logical outcome would cause a catastrophic failure (a collision).

Required Modifications & Additions:

To handle simultaneous emergencies, the knowledge base must break ties by introducing conflict resolution logic. We need facts that represent spatial/temporal data (`Location`, `Distance`, `Priority Level`) and rules to evaluate them.

1. Priority Handling Rule: Higher severity emergencies take precedence.

$\forall x, y: \text{EmergencyVehicle}(x) \wedge \text{EmergencyVehicle}(y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{ClearPath}(x) \wedge \text{Wait}(y)$

2. Distance-Based Resolution: If priorities are equal, the closer vehicle proceeds first.

$\forall x, y, \text{loc}: \text{Approaching}(x, \text{loc}) \wedge \text{Approaching}(y, \text{loc}) \wedge \text{CloserTo}(x, \text{loc}, y) \rightarrow \text{GrantSignal}(x) \wedge \text{HoldSignal}(y)$

3. Intersection Lock Rule: Ensure mutually exclusive green lights for perpendicular paths.

$\forall \text{path1}, \text{path2}: \text{Perpendicular}(\text{path1}, \text{path2}) \rightarrow \neg(\text{GreenSignal}(\text{path1}) \wedge \text{GreenSignal}(\text{path2}))$

Case Study 3: Agricultural Expert Reasoning System

(a) Conversion to Conjunctive Normal Form (CNF)

CNF requires statements to be formulated as an AND of ORs (a conjunction of clauses, where each clause is a disjunction of literals).

Rule P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

- Eliminate implication: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$ (CNF 1)

Rule P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

- Eliminate implication: $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

- Apply De Morgan's Law: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$ (CNF 2)

Rule P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

- Eliminate implication: $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

- Apply De Morgan's Law: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$ (CNF 3)

Facts:

- SoilDry (CNF 4)

- CropWheat (CNF 5)

(b) Resolution Refutation Proof

Goal to prove: ApplyDripMethod . To use resolution refutation, we negate the goal: $\neg \text{ApplyDripMethod}$, and add it to our list of CNF clauses to seek a contradiction.

List of Clauses:

C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$ (Not needed for this proof)

C4: SoilDry

C5: CropWheat

C6: $\neg \text{ApplyDripMethod}$ (Negated Goal)

Resolution Steps:

1. Resolve C6 ($\neg \text{ApplyDripMethod}$) with C2 ($\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$).

- The term ApplyDripMethod cancels out.
- Resolvent R1: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat}$

2. Resolve R1 ($\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat}$) with C5 (CropWheat).

- The term CropWheat cancels out.
- Resolvent R2: $\neg \text{IrrigationNeeded}$

3. Resolve R2 ($\neg \text{IrrigationNeeded}$) with C1 ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$).

- The term IrrigationNeeded cancels out.
- Resolvent R3: $\neg \text{SoilDry}$

4. Resolve R3 ($\neg \text{SoilDry}$) with C4 (SoilDry).

- The terms cancel out completely.
- Resolvent R4: Empty Clause ($\square$)

The derivation of the empty clause confirms that our negated goal creates a logical contradiction. Therefore, the original theorem ' ApplyDripMethod ' is true.

(c) Knowledge Base Modification Analysis

Scenario: The fact ' $\text{SoilDry} = \text{True}$ ' is removed from the Knowledge Base. We must analyze how this impacts the derivation of ' CropAtRisk '.

Step-by-step reasoning:

1. Without ' SoilDry ', Rule P1 ($\text{SoilDry} \rightarrow \text{IrrigationNeeded}$) cannot be triggered via Modus Ponens. We can no longer deduce ' IrrigationNeeded '.
2. Since ' IrrigationNeeded ' is unproven, the antecedent of Rule P2 ($\text{IrrigationNeeded} \wedge \text{CropWheat}$) is false/unknown. Consequently, we cannot derive ' ApplyDripMethod '.
3. Now, we evaluate Rule P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$.
4. In strictly classical propositional logic without the Closed World Assumption (CWA), the inability to prove ' ApplyDripMethod ' does NOT mean that ' $\neg \text{ApplyDripMethod}$ ' is explicitly true. It simply remains unknown. Because ' $\neg \text{ApplyDripMethod}$ ' is not established as a known fact, the antecedent for Rule P3 is not satisfied.

Conclusion: ' CropAtRisk ' does NOT logically follow.

Note on Non-Monotonic Reasoning / CWA: If the system employs the Closed World Assumption (where anything not known to be true is assumed false), then failing to derive 'ApplyDripMethod' means ' $\neg$ ApplyDripMethod' is assumed True. In that specific paradigm, Rule P3 would trigger (since CropWheat is True), and 'CropAtRisk' would follow. However, based purely on standard deduction provided in the case, it does not follow.

Case Study 4: Wumpus World Knowledge Agent

(a) Belief State & Modus Ponens Derivations

The agent's initial percepts are translated into facts in the Knowledge Base:

F1: Stench[1,2]

F2: Breeze[1,1]

F3: Glitter[2,2]

Applying Modus Ponens systematically to the KB rules:

1. From F1 (Stench[1,2]) and Rule 1 (Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]):

- Derived Conclusion 1: Wumpus is at [1,1] OR [1,3] OR [2,2]. ($W[1,1] \vee W[1,3] \vee W[2,2]$)

2. From F2 (Breeze[1,1]) and Rule 2 (Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]):

- Derived Conclusion 2: Pit is at [1,2] OR [2,1]. ($P[1,2] \vee P[2,1]$)

3. From F3 (Glitter[2,2]) and Rule 3 (Glitter[x,y] $\rightarrow$ Gold is at [x,y]):

- Substitution: {x/2, y/2}

- Derived Conclusion 3: Gold is at [2,2].

(b) Forward Chaining for Entity Localization

Using the derived knowledge, the agent uses forward chaining to further isolate the threats.

Wumpus Location Reasoning:

We derived $W[1,1] \vee W[1,3] \vee W[2,2]$. In a standard Wumpus World environment, the agent starts at [1,1]. If the Wumpus were at [1,1], the agent would be dead, or it violates standard axioms (Agent starts in a safe square). Assuming the starting square [1,1] is known to be free of a Wumpus ($\neg W[1,1]$), our disjunction reduces to $W[1,3] \vee W[2,2]$. Without further exploration or percepts (like the absence of Stench in a neighboring cell), the agent cannot logically pinpoint the exact single cell containing the Wumpus. Possible locations remain [1,3] and [2,2].

Pit Location Reasoning:

We derived $P[1,2] \vee P[2,1]$. Based strictly on the single percept at [1,1], there is insufficient data to resolve this disjunction. The Pit could be in either adjacent square. Possible locations remain [1,2] and [2,1].

(c) Path Safety Evaluation

Proposed Path: $[1,1] \rightarrow [1,2] \rightarrow [2,2]$. Goal: Collect the gold at $[2,2]$.

To determine if the path is safe, we must prove $\text{Safe}[x,y]$ for each step using Rule 4:
 $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$.

Step 1: Cell $[1,1]$.

Assuming the agent starts here, it is typically axiomatically safe ($\neg \text{W}[1,1] \wedge \neg \text{P}[1,1]$).

Step 2: Cell $[1,2]$.

To step into $[1,2]$, the agent must prove $\text{Safe}[1,2]$. According to our derivations in part (b), $\text{Pit}[1,2]$ is a possible reality ($\text{P}[1,2] \vee \text{P}[2,1]$). We cannot logically deduce $\neg \text{Pit}[1,2]$ from our current knowledge base. Because we cannot prove $\neg \text{Pit}[1,2]$, we cannot satisfy the antecedent of Rule 4. Thus, $\text{Safe}[1,2]$ remains unknown.

Conclusion:

The path is NOT logically safe. Moving into $[1,2]$ carries a 50% risk of falling into a Pit based on the agent's current belief state. A purely logical agent operating on strict deduction (requiring absolute certainty of safety before moving) would refuse to take this path. It must explore $[2,1]$ or other accessible nodes first to gather more percepts (e.g., lack of breeze in $[2,1]$) to resolve the pit's location.